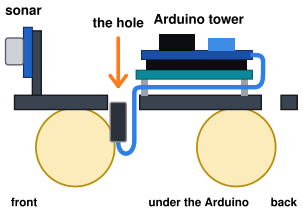
How to plug in Bluetooth

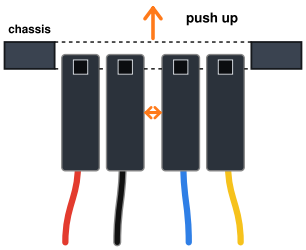
Then drive with Sticks, or program with Blocks

1. Find the hole



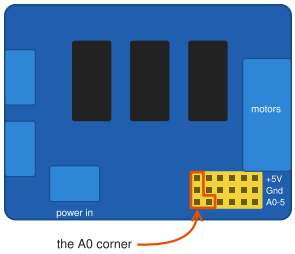
Look from the side. The hole is in the chassis **between the sonar sensor bracket and the Arduino tower**. The four wires come from the back of the tower, run **under the Arduino**, dip **a short way under the robot** and come up into it.

2. Wiggle and push



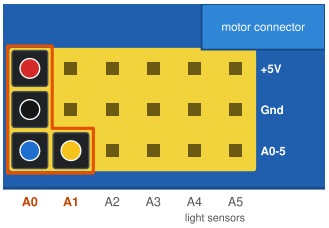
Push the connectors up into the hole **from below**. It is **just big enough for 4**, side by side: wiggle and push. Push them **about halfway in**, then stop pushing. **Tip:** turning each connector a quarter turn (90 degrees) may help them fit.

3. Find the analog header



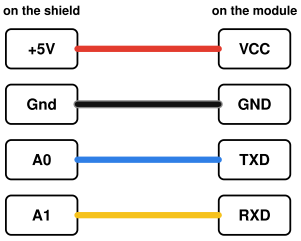
The other ends of the wires are on the motor shield, on the analog header: three rows of pins at the edge. The **top row is all +5V**, the **middle row is all Gnd**, the **bottom row is A0 to A5**.

4. Make an L



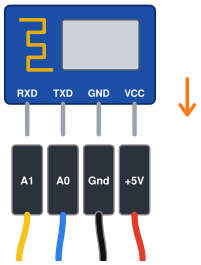
The four wires make an **L**: **+5V**, **Gnd** and **A0** straight down the A0 column, and **A1** next to A0. Check yours looks like this.

5. Which wire is which



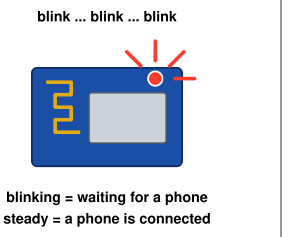
Follow each wire from the shield to the hole. **+5V** goes to **VCC**, **Gnd** to **GND**, **A0** to **TXD**, **A1** to **RXD**. Your wires can be any colors.

6. Plug in the module



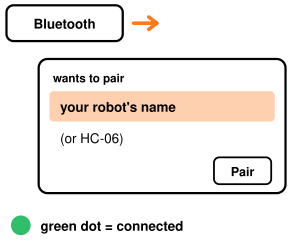
Push the module's pins into the connectors. **Read the names printed on the module**: each pin goes into the wire from step 5.

7. Switch on



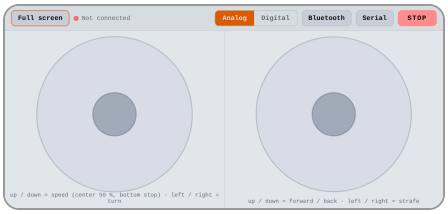
Switch the robot on. The light on the module **blinks**: it is waiting for you. No light? Check VCC and GND.

8. Connect



Press **Bluetooth**. Tap **your robot's name** (it may be listed as **HC-06**), then **Pair**. Android or a computer, in Chrome. Not iPhones.

9. Drive it: Sticks

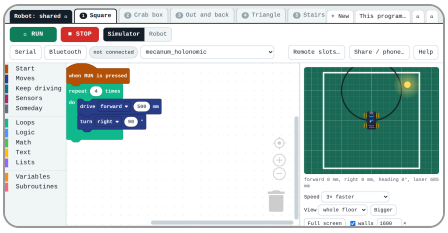


Sticks

robotics.makerlabkids.com/mecanum/sticks

Turn the phone sideways and press Full screen. **Right stick:** which way to go. **Left stick:** up is faster, down is slower, left and right turn. Lift your thumbs to stop.

10. Program it: Blocks



Blocks

robotics.makerlabkids.com/mecanum/blocks

Snap blocks together under **when RUN is pressed**. Press **Bluetooth**, pick your robot, choose **Robot**, then press **RUN**. Best on a computer or a tablet.